- The value of the discriminant $(\Delta = b^2 - 4ac)$ decides the nature of roots as follows
  - (i) When $\Delta > 0$, the roots are real and unequal.
  - (ii) When $\Delta = 0$, the roots are real and equal.
  - (iii) When $\Delta < 0$, there are no real roots.
- Solving quadratic equation graphically.
- A matrix is a rectangular array of elements arranged in rows and columns.
- Order of a matrix

  If a matrix A has m number of rows and n number of columns, then the order of the matrix A is (Number of rows) $\times$ (Number of columns) that is, $m \times n$ .We read $m \times n$ as m cross n or m by n. It may be noted that $m \times n$ is not a product of m and n.
- Types of matrices
  - (i) A matrix is said to be a row matrix if it has only one row and any number of columns. A row matrix is also called as a row vector.
  - (ii) A matrix is said to be a column matrix if it has only one column and any number of rows. It is also called as a column vector.
  - (iii) A matrix in which the number of rows is equal to the number of columns is called a square matrix.
  - (iv) A matrix is said to be a zero matrix or null matrix if all its elements are zero.
  - (v) If A is a matrix, the matrix obtained by interchanging the rows and columns of A is called its transpose and is denoted by A^T .
  - (vi) A square matrix, all of whose elements, except those in the leading diagonal are zero is called a diagonal matrix.
  - (vii) A diagonal matrix in which all the leading diagonal elements are same is called a scalar matrix.
  - (viii) A square matrix in which elements in the leading diagonal are all "1" and rest are all zero is called an identity matrix (or) unit matrix.
  - (ix) A square matrix in which all the entries above the leading diagonal are zero is called a lower triangular matrix.

    If all the entries below the leading diagonal are zero, then it is called an upper triangular matrix.
  - (x) Two matrices A and B are said to be equal if and only if they have the same order and each element of matrix A is equal to the corresponding element of matrix B. That is, $a_{ij} = b_{ij}$ for all i, j.

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